

# Photo-Synth 2

*Play the colours of four photographs.*

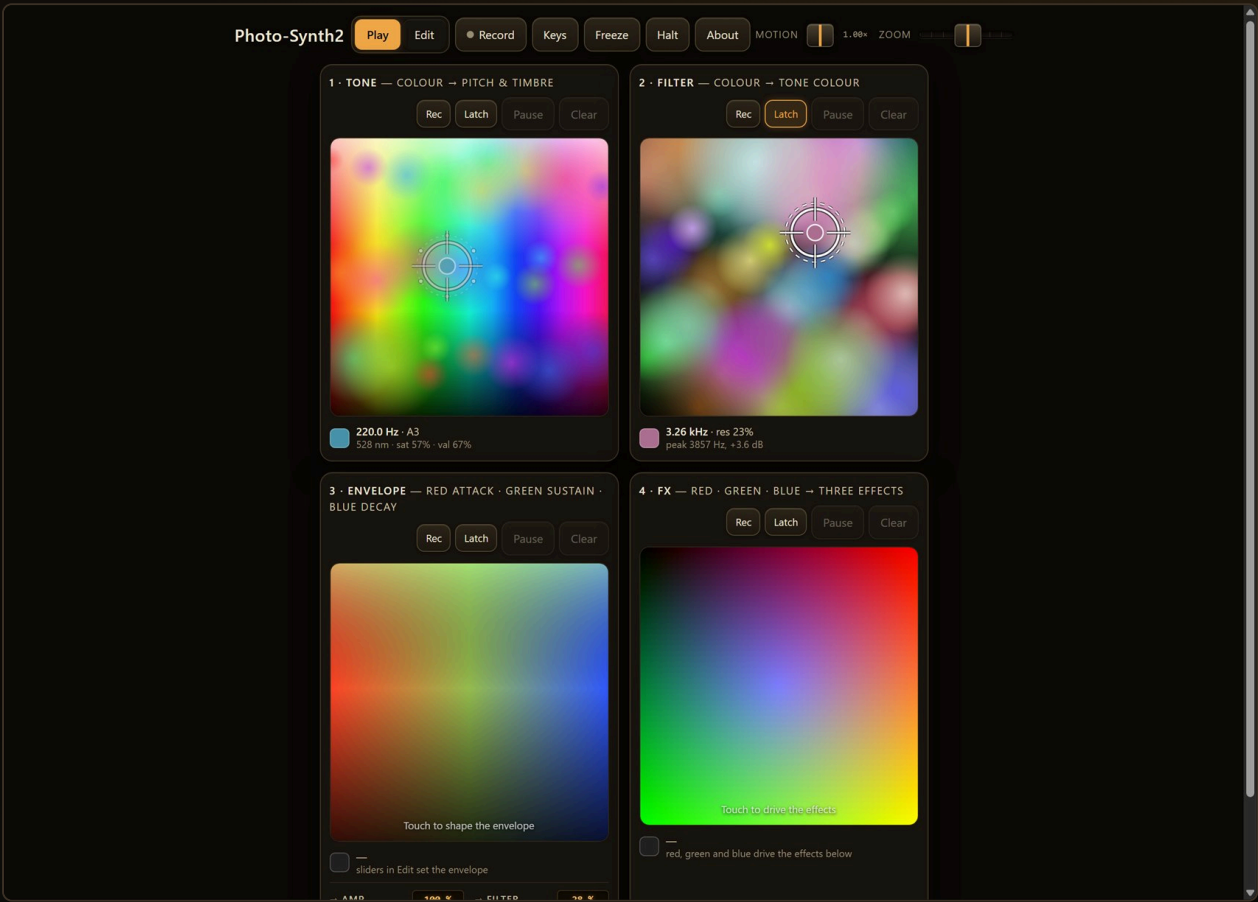


Photo-Synth 2 turns photographs into a playable synthesiser. You do not edit sound with numbers; you drag a finger — or a mouse — across a picture, and the colour under the cursor becomes the sound.

There are four photo pads, and each one listens to colour in a different way. The **first** is the oscillator: hue chooses the pitch, saturation morphs the waveform, brightness sets the loudness. The **second** is the filter. The **third** is the envelope. The **fourth** drives three effect parameters of your choosing. All four work at once, and each can be latched in place or set looping along a recorded gesture, so the instrument keeps moving on its own while you play something else.

Underneath the pictures is a complete virtual-analog voice — two band-limited oscillators that drift a few cents apart, a zero-delay-feedback ladder filter with a saturating core, a sixteen-voice pool, and a seven-module stereo effects rack you can reorder. All of it runs in native C++ inside your DAW, sample-accurate on host MIDI, and every control is automatable.

## How the four pads divide the work

| PAD          | WHAT THE COLOUR UNDER THE CURSOR DOES                                                                                                                                                   |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 · Tone     | Hue → pitch (red is your base note, violet the top of the range). Saturation → waveform, from a pale sine to a full saw or pulse. Green → detune and vocal edge. Brightness → loudness. |
| 2 · Filter   | Brightness → cut-off, from 80 Hz in the shadows to 14 kHz in the highlights. Saturation → resonance. Hue → a resonant peak swept from 180 Hz to 6 kHz.                                  |
| 3 · Envelope | Red → attack (from a 1.5 s swell down to 2 ms). Green → sustain level. Blue → decay and release (30 ms up to a 4 s tail).                                                               |
| 4 · FX       | The red, green and blue strengths each drive one effect parameter, chosen in the menus under the pad. Black is all three off, white is all three full.                                  |

Above all that sits a shelf of **Spectra** — six characters that do not process the sound so much as re-decide how every voice is built: a drone engine, a liquid swirl, a bit-crushed industrial machine, a cathedral, a haunted tape and a swarm of insects. Arm any combination; each hands back exactly what it changed when you switch it off.

**Nothing leaves your machine.** Photos are decoded and sampled locally, audio is synthesised in the plugin. There is no network access of any kind.

## Installing

- Unzip the download and copy the whole `Photo-Synth2.vst3` folder into `C:\Program Files\Common Files\VST3\`.

- Rescan plugins in your DAW. The instrument appears as **Photo-Synth2** by Brokild.
- Windows 10/11, 64-bit. The editor uses the Microsoft Edge WebView2 runtime, which is already present on current Windows installations.

A new instance opens on **Dreamy Hoover** — a stacked, gliding, pentatonic pad with a cursor already resting on the first photo. It waits in silence: the cursor is placed but not latched, so the patch is ready to play rather than already playing. Touch photo 1, or press a key, and there it is. Reach for **Naked Oscillator** in the Presets panel when you want the bare engine with nothing between the photo and your ears.

Load it, drag on the first photo, and you are playing. Everything below the first two steps is refinement.

## 1 Add the instrument to a track.

It arrives with four demo photos already loaded, so it makes sound immediately — you do not need to import anything to start.

## 2 Drag slowly across photo 1.

The note follows the colour: warm reds are low, cool blues and violets are high. Lift the cursor and the note releases. This is the whole instrument in one gesture.

## 3 Rest the cursor somewhere on photo 2.

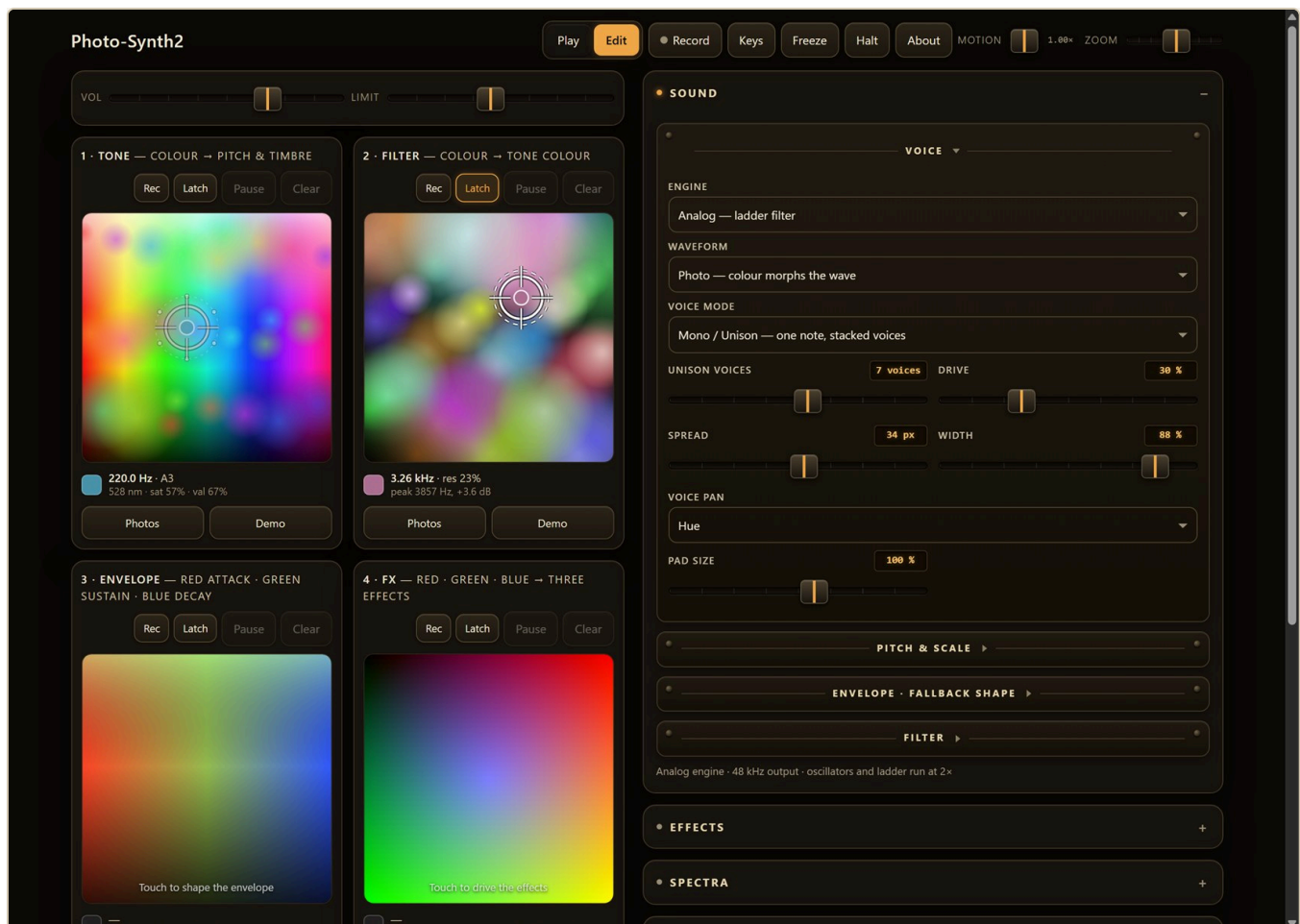
A dark patch mutes the tone, a bright patch opens it up, and a saturated colour adds resonance. Photo 2 keeps whatever colour you left it on — it is a setting, not a gesture.

## 4 Press *Keys* in the top bar and play the keyboard.

A key sets the **base note**. Everything the photo does is an interval above it, so the same picture can be played in any key. Chords work in Polyphonic voice mode, and MIDI from your DAW plays it too.

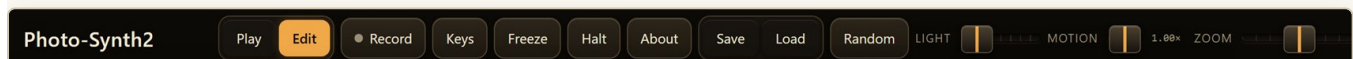
## 5 Switch to *Edit* and open *Sound*.

Here are the voice, the scale, the envelope and the filter type. Then open **Effects** for the pedal rack. Play while you change things; nothing needs to be stopped.



**Edit mode.** The photos and the analysis strip stay on the left; every panel lives in the right-hand column. **Play** mode hides all of it and leaves only the photos — that is the performance view.

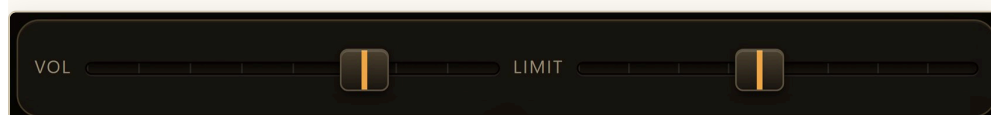
## The top bar



From the left: the Play / Edit switch, audio **Record**, the on-screen **Keys**, **Freeze**, **Halt**, **About**, then console **Light**, **Motion** speed and the interface **Zoom**.

| CONTROL            | WHAT IT DOES                                                                                                                                                                                                                                                                                                   |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Play / Edit</b> | Play shows only the four photos. Edit reveals the transport, the analysis strip and all panels.                                                                                                                                                                                                                |
| <b>Record</b>      | Captures the plugin's own output. <b>Save WAV</b> in the File panel writes it as 24-bit stereo. Maximum 180 seconds.                                                                                                                                                                                           |
| <b>Keys</b>        | Shows or hides the on-screen keyboard docked at the bottom.                                                                                                                                                                                                                                                    |
| <b>Freeze</b>      | Holds the delay tail forever: the loop's input closes, feedback goes to unity under a soft limiter, damping opens wide. Play over the frozen bed.                                                                                                                                                              |
| <b>Halt</b>        | A tape stop. Pitch spins down to a halt; press again to spin back up.                                                                                                                                                                                                                                          |
| <b>Light</b>       | Lifts the whole console out of the dark — panels, pads, buttons and the page behind them — without touching the photographs, which keep their own contrast. Off by default; double-click the slider to return there. It is remembered for this installation, like Zoom, rather than saved with a patch.        |
| <b>Save</b>        | Writes the whole patch — controls, chain, cursors, paths and small copies of the photographs — to a file, through an ordinary Save dialog opened on your preset folder. It is in the title bar rather than in a panel so it is reachable in Play mode, when you are least likely to want to go hunting for it. |
| <b>Load</b>        | Opens a menu of everything in the preset folder. Sub-folders fly out sideways as sub-menus and sit alongside the loose presets, so a library can be sorted by kind. <b>Browse...</b> at the foot of the menu reaches anywhere on disk; <b>Preset folder...</b> moves the library.                              |
| <b>Random</b>      | Throws a new set of numbers at every sound control at once. It deliberately leaves the photographs, the cursor positions and the motion paths exactly where they are — so a patch you have choreographed keeps its choreography and only changes its voice. Volume, Limit and Motion are left alone too.       |
| <b>Motion</b>      | How fast every path and recorded gesture replays, from a quarter speed to four times — a master over the per-pad speeds in Automotion. It lives up here rather than in a panel because the panels are hidden in Play mode, which is exactly when a moving cursor matters.                                      |
| <b>Zoom</b>        | Scales the whole console uniformly, for large or small plugin windows.                                                                                                                                                                                                                                         |

## Volume and the limiter

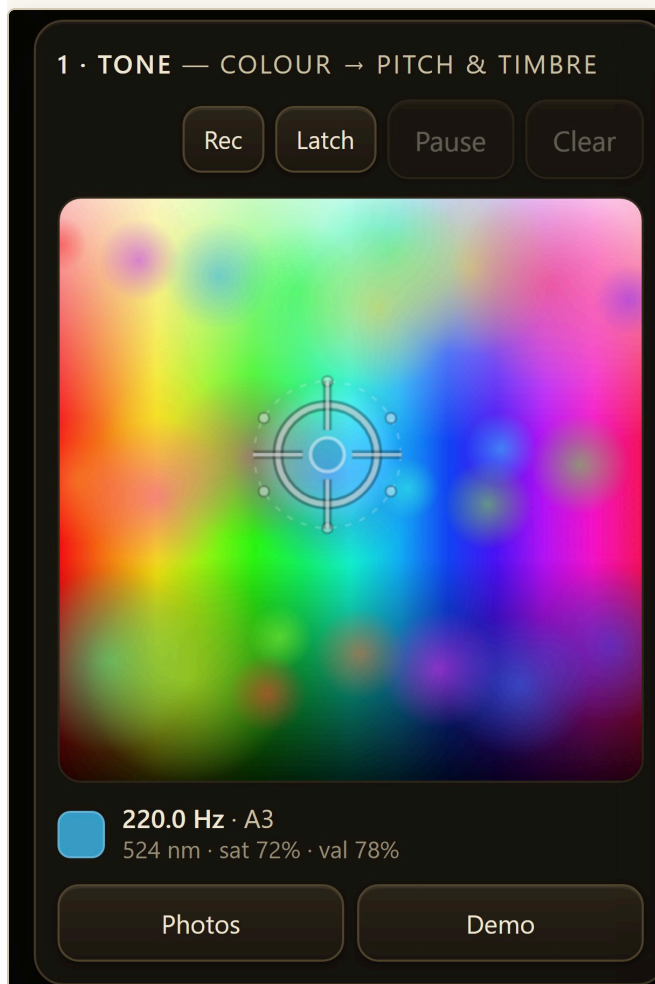


**Vol** is the master level. **Limit** is a transparent peak limiter.

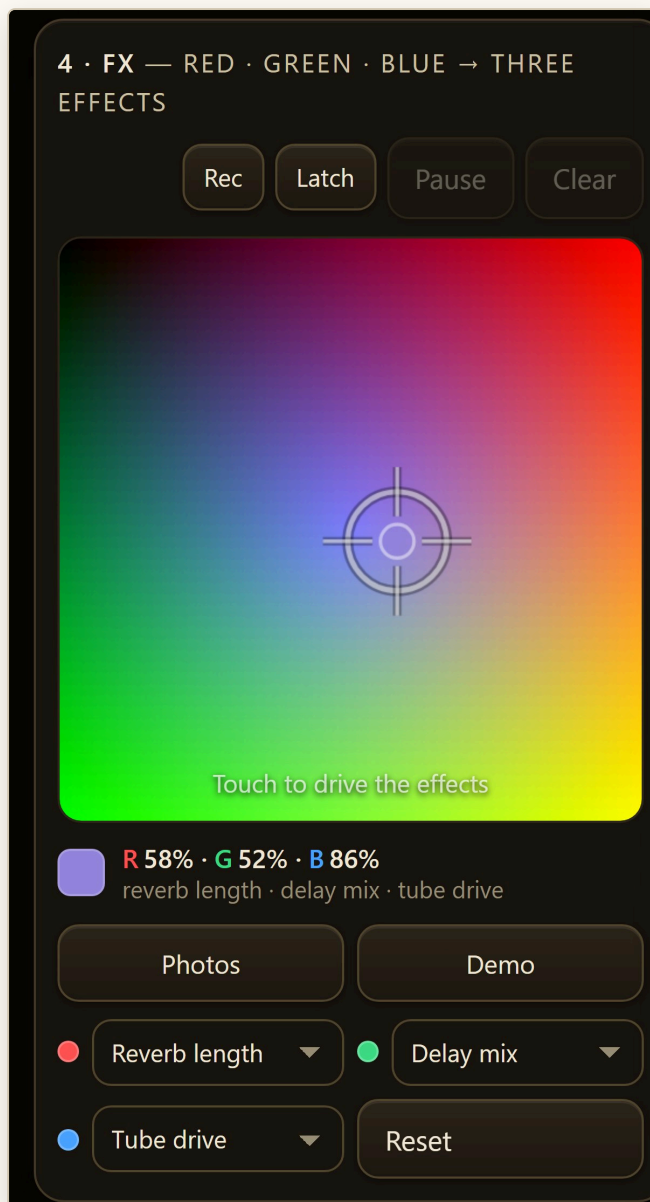
The limiter is deliberately plain: pure downward gain riding above a threshold, no makeup gain and no tone shaping, so it does not colour the sound. It is there because a resonant ladder filter with the envelope sweeping it can produce sudden peaks. At the default 40 % it catches those without ever touching ordinary

playing. Turn it up if you are pushing resonance and drive hard; turn it to zero if you would rather do all the limiting yourself downstream.

Every pad carries the same controls — **Rec**, **Latch**, **Pause** and **Clear** along the top, **Photos** and **Demo** underneath — and differs only in what it does with the colour it samples.



**Pad 1 · Tone.** While you drag, a magnifier shows the pixels under the cursor, and the readout gives the exact frequency, the nearest note with its cents offset, the sampled wavelength, saturation and brightness.



**Pad 4 · FX.** Three menus assign what red, green and blue each drive. Out of the box: red → reverb length, green → delay mix, blue → tube drive. **Reset** returns those three sliders to their defaults and releases the pad.

## Pad controls

| CONTROL       | WHAT IT DOES                                                                                                                                                                                                                               |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Photos</b> | Loads an image from disk (JPEG, PNG, and anything else Windows can decode). The photo is stored inside the plugin, so it travels with your project and your presets.                                                                       |
| <b>Demo</b>   | Loads the pad's generated demo image — a hue sweep for Tone, soft colour blobs for Filter, a red-to-blue ramp for Envelope, and an RGB field for FX.                                                                                       |
| <b>Latch</b>  | Leaves the cursor where you lifted it. On Tone the note keeps sounding; on the other pads the setting is held. Tap the pad again to move it, or switch Latch off to release.                                                               |
| <b>Rec</b>    | Arms motion recording for that pad alone — see below.                                                                                                                                                                                      |
| <b>Pause</b>  | Holds a replaying gesture exactly where it stands, without touching the recording. Release it and the loop carries on from that point rather than jumping — it is credited the time it stood still. Greyed out until the pad has a motion. |
| <b>Clear</b>  | Throws the recorded gesture away. The cursor stays where it was.                                                                                                                                                                           |

## Motion: gestures that loop

# 1 · TONE — COLOUR → PITCH & TIMBRE

AUTO

Rec

Latch

Pause

Clear



196.0 Hz · G3

545 nm · sat 91% · val 75%

Photos

Demo

A recorded gesture, held with **Pause**. The dashed line is the path, and the **white dot marks where the loop begins** — aim for it as you finish drawing and the gesture closes into a seamless loop.

**AUTO** lights whenever a pad has one armed.

Press **Rec** on the pad you want to automate, then draw. Release, and the gesture starts looping immediately — the cursor retraces your path, sampling colour as it goes, at the speed set by **Motion** in the top bar. Every new note re-phases all running motions, so a phrase lands the same way each time. Dragging an automated cursor takes manual control without erasing the loop; a single tap starts it again.

**Pause** stops the cursor where it stands and leaves everything else alone: the path is still armed, still drawn, and releasing the button picks the loop up where it left off instead of snapping to a new phase. It is the

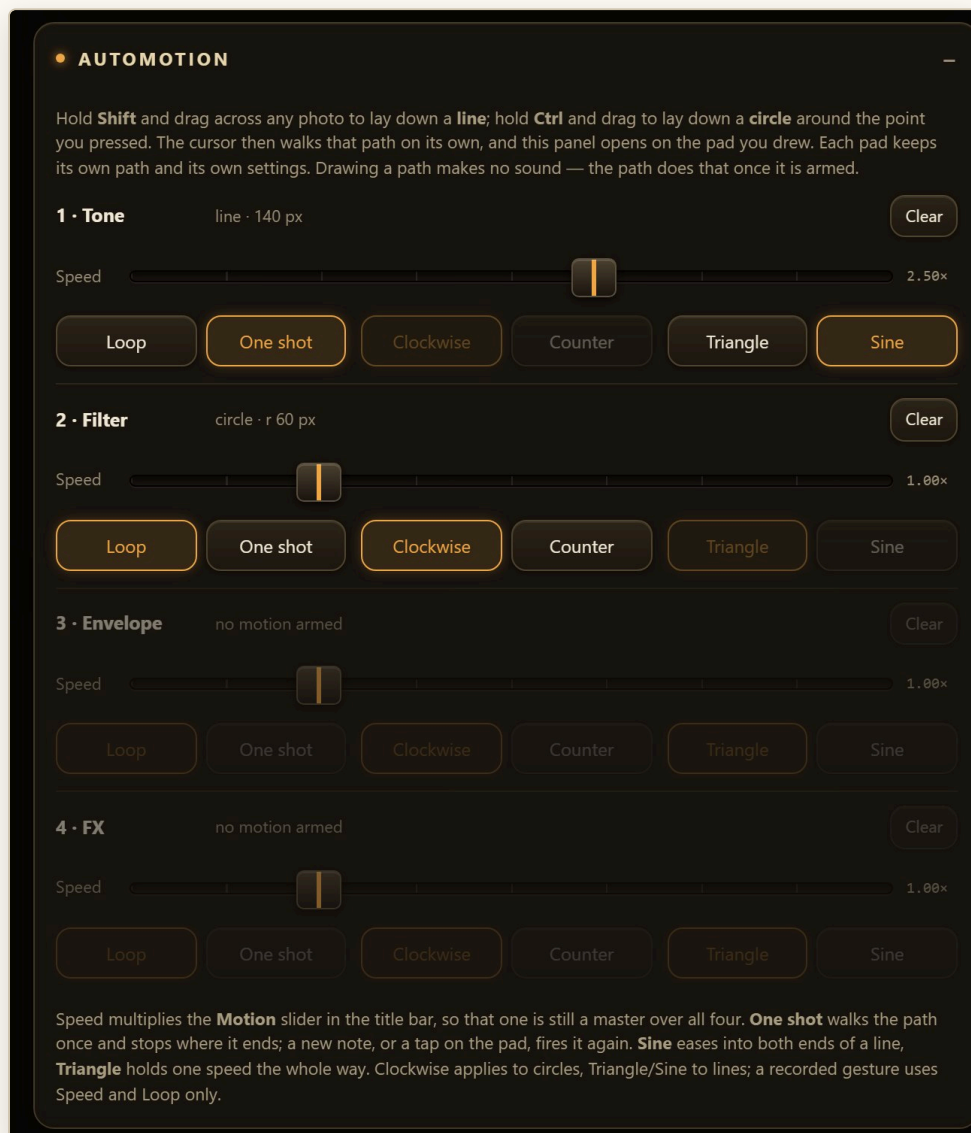
difference between freezing a texture for a bar and losing the gesture you spent a minute getting right. **Clear** is the one that discards it.

## Automotion: lines and circles you draw in one gesture

Recording a gesture is the free-hand way. The other way needs no arming at all: hold a modifier and drag straight across the photograph.

|                     |                                                                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Shift + drag</b> | Lays down a <b>line</b> from where you pressed to where you let go. The cursor walks up and down it.                                         |
| <b>Ctrl + drag</b>  | Lays down a <b>circle</b> centred on the point you pressed, with the drag setting the radius. A small cross marks the centre while you draw. |

Drawing a path makes no sound — you are describing a shape, not performing one — and the moment you release, the path arms and starts walking. The panel opens on the pad you drew, everything else folds away, and the row for that pad flashes.



One row per photo pad. A control that cannot apply goes grey rather than disappearing, so the row never moves: direction is a circle idea, travel shape a line idea, and a recorded gesture uses Speed and Loop only.

| CONTROL                    | WHAT IT DOES                                                                                                                                                                                                                                                                                         |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Speed</b>               | 0.1× to 4× for this pad alone, multiplied by <b>Motion</b> in the title bar. Safe to move while the path is running: the replay carries an accumulated phase rather than a start time, so the cursor never jumps to catch up with a new formula.                                                     |
| <b>Loop / One shot</b>     | Loop walks the path for ever. One shot walks it once and stops where it ends, leaving the cursor — and the colour under it — exactly there. The path stays armed: the next note, or a tap on the pad, fires it again. That turns a one-shot line into a sweep you play rather than a loop you fight. |
| <b>Clockwise / Counter</b> | Which way a circle turns. Switching it reverses the cursor in place rather than restarting the circle.                                                                                                                                                                                               |
| <b>Triangle / Sine</b>     | How a line travels between its ends. Triangle holds one speed the whole way and turns sharply at each end. Sine eases into both ends and is quickest through the middle — the difference between a scan and a breath.                                                                                |
| <b>Clear</b>               | Discards the path for that pad. The same button sits on the pad itself.                                                                                                                                                                                                                              |

Every pad keeps its own path and its own four settings, and all of it travels in the preset and in the DAW project along with everything else.

**Live camera is not part of the plugin.** A DAW plugin has no business opening your webcam, so the browser version's camera capture and live feed were left out. Everything else from the original is here.

## The envelope photo

3 · ENVELOPE — RED ATTACK · GREEN  
SUSTAIN · BLUE DECAY

Rec

Latch

Pause

Clear



Touch to shape the envelope



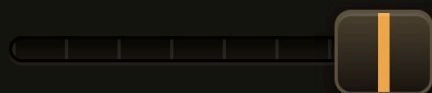
39 ms att · 84 ms dec  
sustain 58%

→ AMP

100 %

→ FILTER

28 %



Photos

Demo

**Pad 3 · Envelope.** Red sets the attack, green the sustain level, blue the decay and release, and the two sliders beneath decide where that shape is sent. Untouched, the fallback sliders in *Sound* → *Envelope* rule the shape instead.

→ **Amp** and → **Filter** appear under this photo, because this is where the envelope is made — and unlike the panels they stay reachable in Play mode. The same pair also appears in *Sound* → *Envelope* beside the fallback shape; they are two faces of one control, so whichever you move, the other follows. → **Amp** is how much of the shape goes to loudness: below 100 % the level blends toward a plain click-free gate, organ-like, so the envelope can be spent on the filter instead. → **Filter** sweeps the cut-off with the envelope, up to five octaves at full — closed when the envelope is down, landing exactly on your filter setting at its peak. Both apply to whichever envelope is in force, the photo's or the sliders'.

This is the pad that changes how the instrument *feels* rather than how it sounds. A red corner gives you a percussive pluck; a deep blue gives a long ringing tail; a bright green area holds notes up indefinitely. Because it is a photo like any other, you can latch a spot or loop a gesture across it and let the articulation drift while you play.

## Seeing what the extra voices read

## 1 · TONE — COLOUR → PITCH & TIMBRE

Rec

Latch

Pause

Clear



220.0 Hz · A3

524 nm · sat 72% · val 78%

Photos

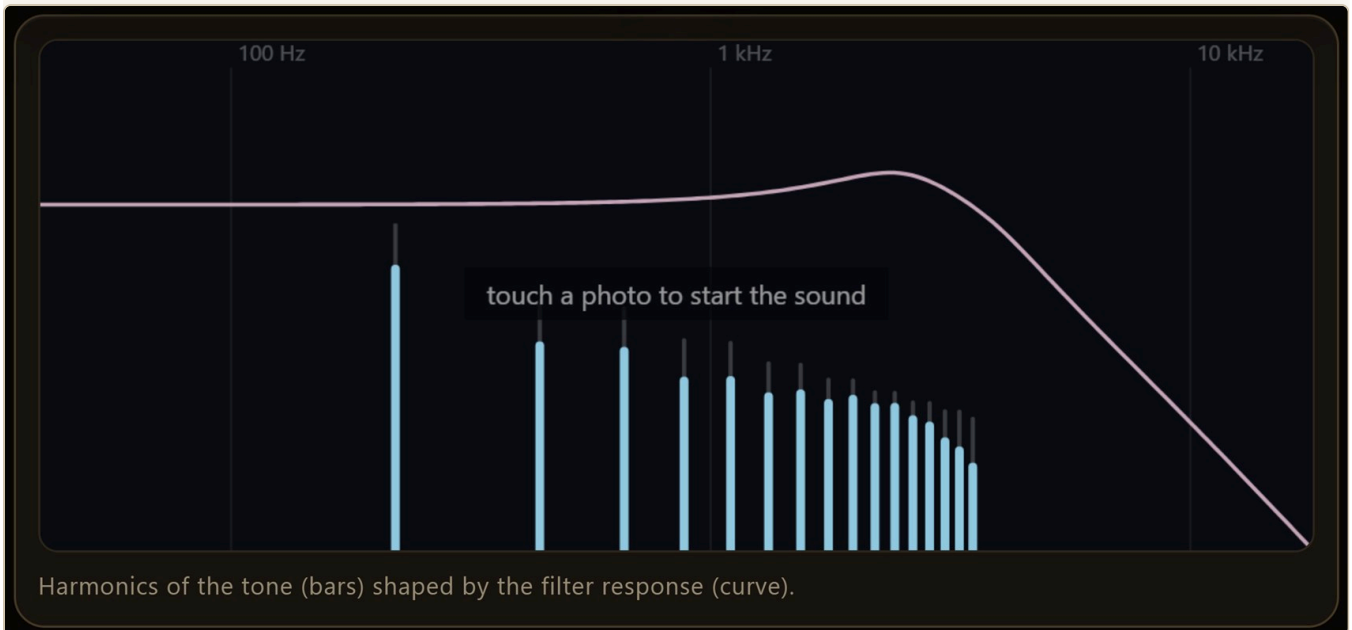
Demo

Twelve voices at a Spread of 46 px: the halo is the patch of photo in play, and every dot is a point that feeds a voice.

Stack more than one voice and they stop reading a single pixel. Voice one plays the colour directly under the cursor; the others sample a ring around it at the **Spread** radius, so the local texture of the photograph becomes a chord. Past seven voices a second, wider ring appears at 1.9× the radius. In **Polyphonic** mode it is each note's oscillators that spread, at 1.0, 1.3 and 1.6× instead.

The tone pad shows you this directly: a soft brightening out to the widest radius in play, a dashed hairline at the Spread radius itself, and a dot on every point that is actually being sampled. Move the Spread slider — now up to 60 px — and you can see the patch of picture the chord is being drawn from. It is the surest way to understand why one part of a photo sounds rich and another sounds flat.

## The analysis strip



Bars are the harmonics of the tone you are playing; the curve is the filter your second cursor is describing. Faint bars are the raw harmonic, solid bars are what survives the filter.

Four engraved units: the voice itself, how colour maps to pitch, the fallback envelope, and the filter. Each one folds away — click a nameplate to close it, and click the folded strip anywhere to open it again — so you can keep only the section you are working on in view.



All four folded. A closed unit is nothing but its nameplate, so there is no small target to hit.

VOICE ▼

ENGINE

Analog — ladder filter ▼

WAVEFORM

Photo — colour morphs the wave ▼

VOICE MODE

Mono / Unison — one note, stacked voices ▼

UNISON VOICES

7 voices

DRIVE

30 %

SPREAD

34 px

WIDTH

88 %

VOICE PAN

Hue ▼

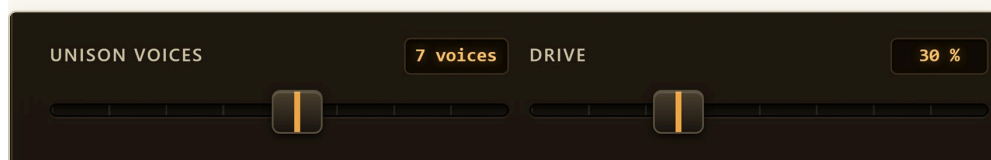
PAD SIZE

100 %

The **Voice** and **Pitch & scale** units.

## Voice

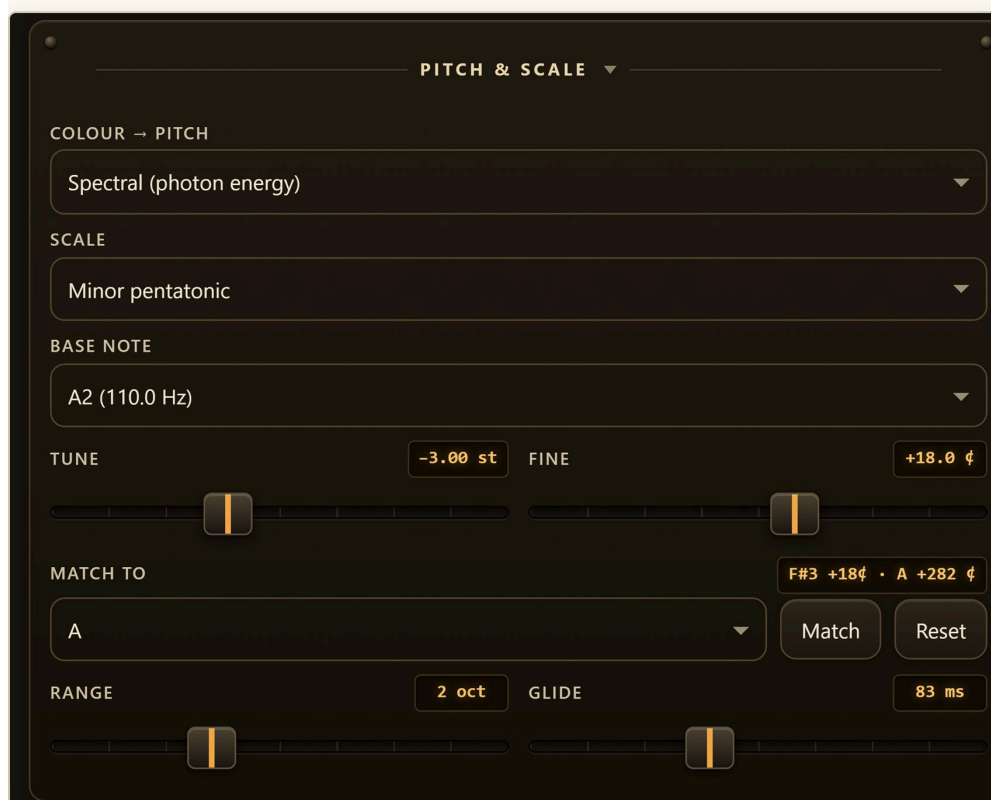
| CONTROL              | WHAT IT DOES                                                                                                                                                                                                                                                                                          |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Engine               | <b>Analog</b> is the virtual-analog voice: two band-limited oscillators through a Moog-style ladder filter, running at twice the output rate. <b>Additive</b> is ten pure sine partials and a clean biquad — glassier, simpler.                                                                       |
| Waveform             | <b>Photo</b> lets the picture morph the wave from sine through triangle to saw or pulse, as saturation rises. The other four — <b>Sine, Triangle, Sawtooth, Square</b> — pin both oscillators to that shape, band-limited and alias-free, and the photo then only affects pitch, loudness and detune. |
| Voice mode           | <b>Mono / Unison</b> plays one note with stacked voices. <b>Polyphonic</b> plays up to four notes at once.                                                                                                                                                                                            |
| Unison voices        | 1–12. Extra voices sample the colours in rings around your cursor, so the local texture of the photo becomes a chord riding on one note. Uniform areas thicken into unison; detailed areas scatter into intervals.                                                                                    |
| Notes / Osc per note | Polyphonic mode only: how many simultaneous notes (1–4) and how many photo-sampling oscillators each note carries (1–4). Oldest note is stolen when you exceed the count.                                                                                                                             |
| Drive                | How hard the ladder's saturating core is pushed. Thickens rather than distorts — the level is compensated.                                                                                                                                                                                            |
| Spread               | The radius, in pixels, of the ring the extra voices sample from. Small values give a tight unison, large values a wide scatter.                                                                                                                                                                       |
| Width                | Stereo width. At zero the two channels are identical for true mono compatibility.                                                                                                                                                                                                                     |
| Voice pan            | Whether voices are panned by the <b>hue</b> they sampled or by their horizontal <b>cursor position</b> .                                                                                                                                                                                              |
| Pad size             | The proportions of all four photo pads, 60–130 %. Taller pads give a gesture more room; shorter ones leave more of the window for the panels. This is not the same as <b>Zoom</b> , which scales the whole console including the pads.                                                                |



**Voices** and **Spread** decide how wide a patch of photo one note reads.

## Pitch & scale

| CONTROL        | WHAT IT DOES                                                                                                                                                                                         |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Colour → pitch | <b>Spectral</b> reads hue as a wavelength and follows photon energy, so the intervals feel physical. <b>Hue angle</b> is a plain linear sweep around the colour wheel.                               |
| Scale          | Continuous, Chromatic, Major, Minor pentatonic, Major pentatonic or Whole tone. Anything but Continuous snaps the colour's interval onto the scale, which makes the pads far easier to play in tune. |
| Base note      | The note a pure red pixel plays. Also set by the keyboard, by your computer keys, and by incoming MIDI.                                                                                              |
| Range          | How many octaves the full colour sweep covers, 1 to 4.                                                                                                                                               |
| Glide          | How long the pitch takes to slide when the colour changes. A little glide makes dragging sound like an instrument; none makes it sound like a scanner.                                               |



Tuning: **Tune** and **Fine** shift everything the synth plays, and **Match to** is a tuner that does the arithmetic for you.

## Tuning the whole instrument

**Tune** slides continuously over  $\pm 12$  semitones — no steps of its own, so it can sit between notes — and **Fine** adds  $\pm 100$  cents on top. Both offset every note the instrument plays: the base note, the keyboard, DAW MIDI and an imported MIDI file. They move the base frequency only, so a held note glides to the new pitch rather than retriggering, and you can tune into something while it sounds.

The **Match to** row is a tuner. Its readout is live — the note sounding right now, and how far away the note you picked is, in cents. **Match** then moves **Tune** and **Fine** so the sounding note becomes that one, always by the shortest distance: a G matched to A goes up two semitones, never down ten, and never by whole

octaves. It reads the frequency the instrument is actually synthesising rather than analysing the output, so it stays exact even while a recorded motion sweeps the pitch around — it catches the note as of the moment you press it. Because the colour under the cursor sets the interval, moving the cursor afterwards moves the pitch again: Match tunes the instrument, it does not pin one note. **Reset** returns both sliders to concert pitch.



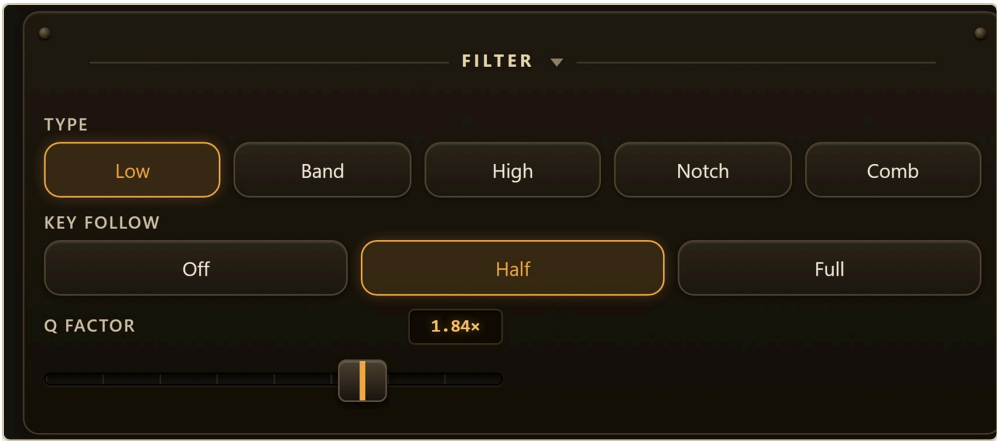
The **Envelope** unit. The drone engine that used to sit beside it now lives in **Spectra**, as Dark Drone Spectrum.

### Envelope · fallback shape

Photo 3 takes the envelope over the moment you touch it, and from then on these two sliders are the fallback. **Shaped by** decides which is in charge: *Photo 3 when it is touched* is the normal behaviour, and *These sliders only* hands control back without having to clear the photo. While the photo is in charge the two sliders grey out and go inert, so you can see at a glance why they are doing nothing.

| CONTROL          | WHAT IT DOES                                                                                                                                                                     |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shaped by        | Photo 3 when touched, or these sliders only.                                                                                                                                     |
| Attack / Release | The basic shape, used whenever photo 3 is not in charge. Without the photo the decay is fixed at 250 ms and the sustain is full, so this is a plain attack-and-release envelope. |
| → Amp / → Filter | The same routing that sits under photo 3, repeated here beside the shape. Move either copy and the other follows.                                                                |
| Legato           | How long the note is held after you lift off photo 1, up to half a second, so tapping from colour to colour glides instead of retriggering every note.                           |

Looking for Dark drone? It has moved, with five new companions, into the **Spectra** panel — see section 05.



The **Filter** unit: the five types, how far the cut-off tracks the keyboard, and the resonance scaler.

## Filter type

| TYPE  | CHARACTER                                                                                                                                                     |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low   | The classic 24 dB ladder low-pass. Dark patches of photo 2 mute the tone.                                                                                     |
| Band  | Keeps a band around the cut-off and discards the rest — thin, vocal, focused.                                                                                 |
| High  | Removes everything below the cut-off. Bright patches become airy and thin.                                                                                    |
| Notch | The inverse of band-pass: sweeps a hollow through the sound instead of a peak.                                                                                |
| Comb  | A tuned feedback delay whose resonances sit at every multiple of the cut-off. With high resonance it rings like a struck string — superb for metallic drones. |

## Key follow

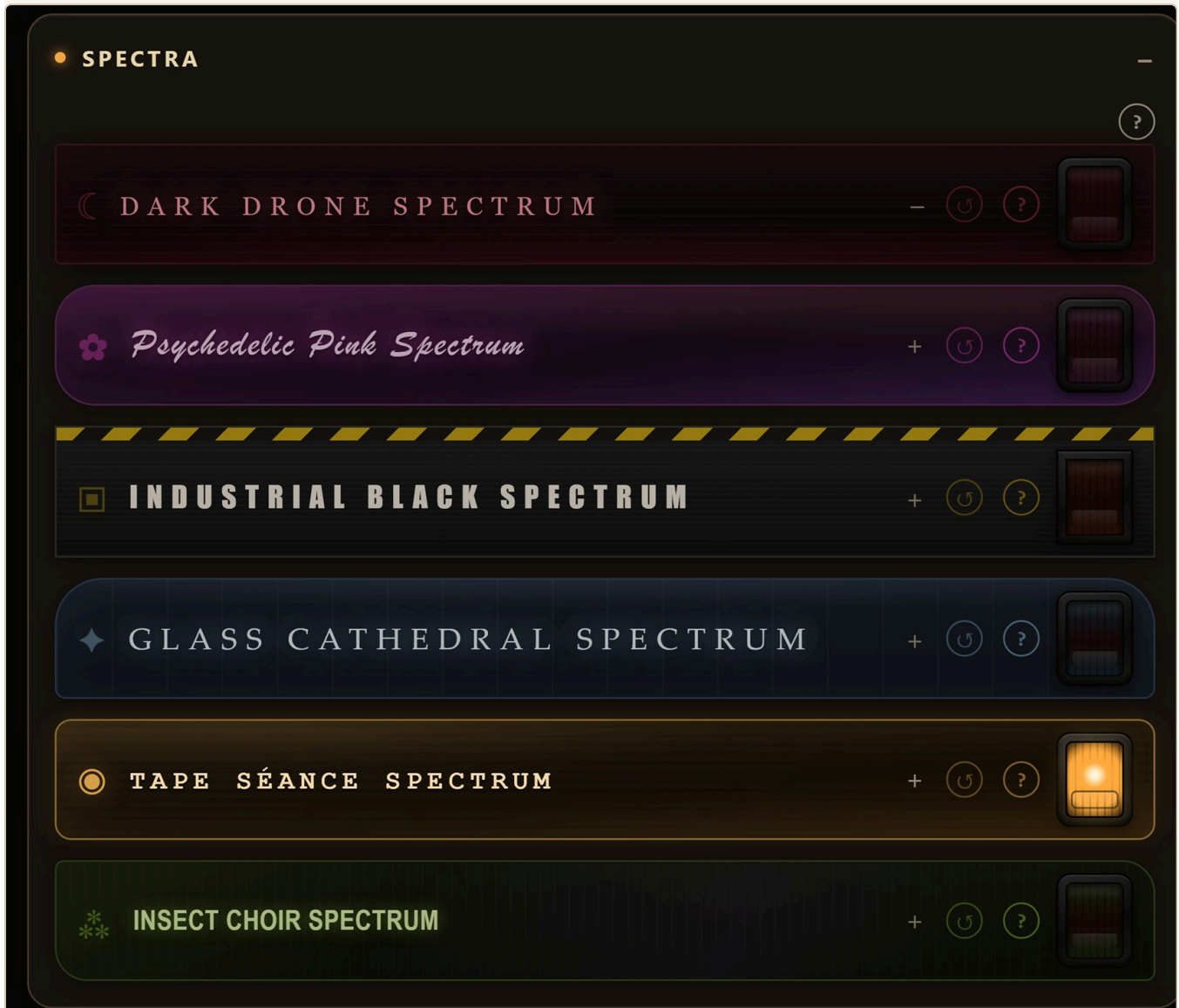
By default the cut-off comes from photo 2 alone, so a bass note and a high one are filtered identically — which makes the low notes sound dull and the high ones thin. **Key follow** ties the cut-off to the keyboard instead: **Full** moves it an octave for every octave you play, **Half** moves it half as far, and **Off** is the old behaviour. It follows the *lowest* note sounding — the bass of a chord rather than whichever key you pressed last — and it measures from C3, so a patch dialled in around there is unchanged and everything above it opens up. Notes from the on-screen keys, your DAW and the poly slots all count.

## Q factor

The photo sets the resonance — saturation in photo 2 pushes it up — and **Q factor** scales what it asks for, from **0.05×** to **4×**, with the centre position leaving the original behaviour exactly as it was. It reaches every mode: the ladder's resonance in Low, Band, High and Notch, the comb's internal feedback, and the biquad Q of the additive engine. Turn it up for singing band-passes and a comb that rings like a string; turn it right down and the resonance all but disappears, which is the setting for a photograph that is making the filter shout, or for a clean sweep with no peak at the corner at all.

On the notch, more resonance deepens and widens the hollow rather than narrowing it — that is the nature of the ladder-derived notch, and the drawn curve in the analysis strip shows you exactly what the filter is doing.

A Spectrum is a character rather than an effect. It does not sit in the signal path waiting for audio; it changes how every voice is built in the first place — which is why each one wears its own enclosure instead of a shared faceplate, and why none of them costs anything at all while its switch is off.



Six characters, one switch each. **Tape Séance** is armed here — the lens lights in the module's own colour.

## Arming, combining, releasing

The illuminated rocker arms a character. You may arm **any combination**: they stack, and where two of them reach for the same control the most recently armed one wins. Switching one off unwinds the stack to that point and re-applies whatever was armed after it, so every release hands back exactly what that character changed and nothing else. Your own chain order, power switches and slider positions survive untouched.

The (?) beside each switch unfolds a description of what that character actually does, and opens the controls if they are collapsed. The panel has one of its own, at the top right, for what a Spectrum is in the first place. Plates stay tight until you ask.

## Whatever you touch is yours

A character takes over a handful of the instrument's controls while it is armed, and hands them back when you release it. But if *you* move one of them in the meantime, it stops being the character's: your value stays where you put it, releasing leaves it alone, and the character will not write over it again. Without that rule an edit made while a character was armed would quietly disappear on release, and the patch would sound different the next time you armed it.

The ⤴ beside the (?) undoes your meddling with that one character: its own knobs go back to the positions they arrive with, and it takes back any of its controls you have borrowed. What the patch looked like *before* you armed it is still remembered, so releasing afterwards still puts that back. Nothing else on the instrument is touched.



Tape Séance Spectrum open, with its explanation unfolded by the (?).

Four of the six switch on effects of their own, and they arrive in whatever order your rack is already in — a character never rearranges your chain. If you want Industrial Black's crush heard by the reverb, put the reverb after the crusher yourself, once, and every character will respect it. Dark Drone and Insect Choir add no effects at all; they work entirely on the voices.



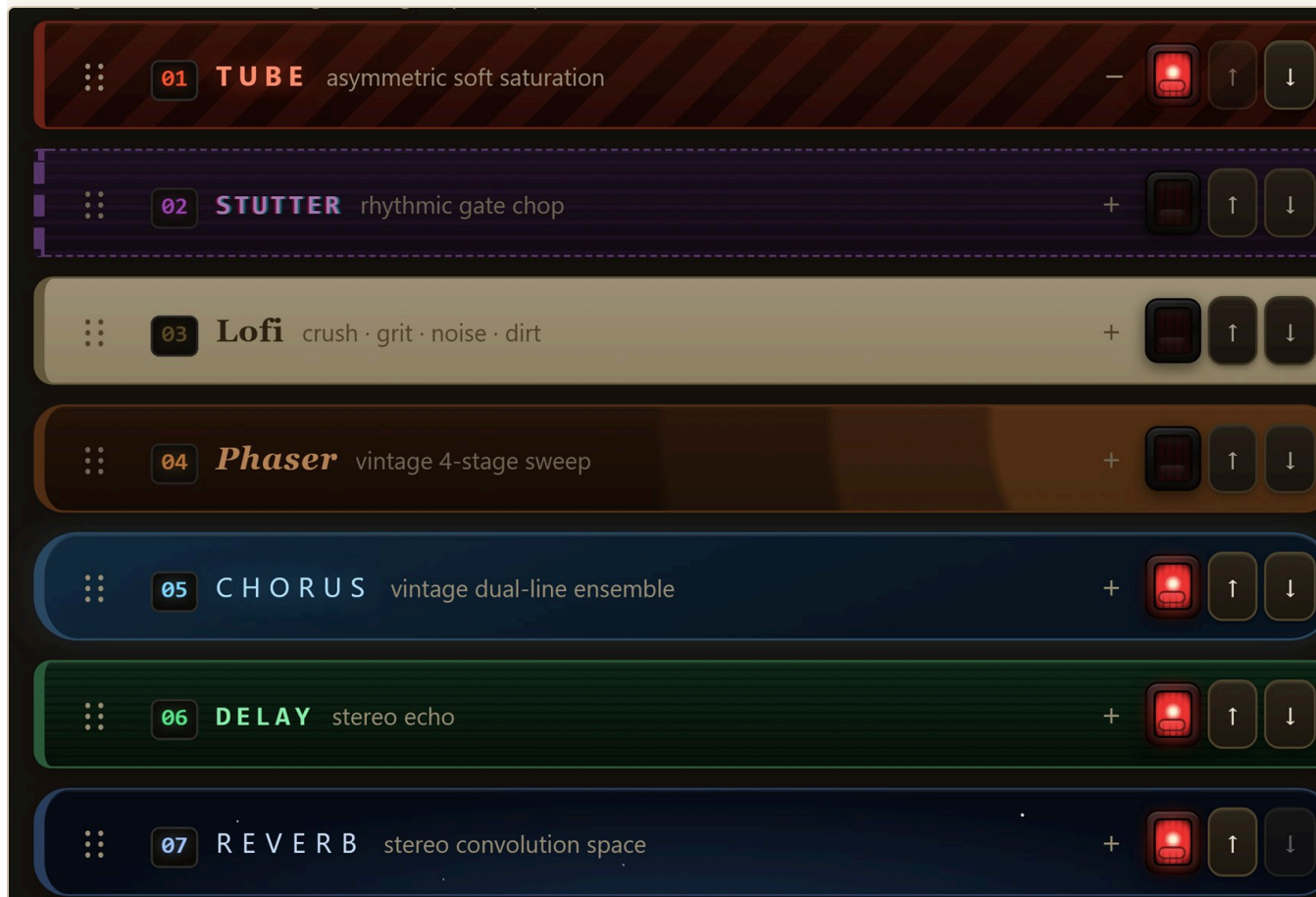
Industrial Black Spectrum, armed. Grind is drive and bit-crush, Chop the gate, Clang the metal.

## The six characters

| CHARACTER                                                                                                                                                                                                                                                                                                                     | WHAT IT DOES TO THE INSTRUMENT                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Dark Drone</b>                                                                                                                                                                                                                                                                                                             | A sub-oscillator under every voice through the same filter, an irregular microtonal cluster instead of tidy unison, an ultra-slow random walk leaning on cut-off, detune and delay time, and notes that sag toward an octave down as they decay. <i>Sub level, Sub octave, Cluster, Env → Pitch, Drift, Drift time.</i>                       |
| <b>Psychedelic Pink</b>                                                                                                                                                                                                                                                                                                       | Everything smeared and slowly rotating: the stereo image swirls, a slow wave bends the tuning, the filter breathes on its own, and the reverb becomes the reverse character so tails swell in rather than die away. <i>Swirl, Smear, Bloom.</i>                                                                                               |
| <b>Industrial Black</b>                                                                                                                                                                                                                                                                                                       | A square wave into the crusher <i>before</i> the filter, so the ladder has to smooth the wreckage; a comb resonance and a tight metal slapback, a stutter on the gate, and notes that stop dead instead of releasing. <i>Grind, Chop, Clang.</i>                                                                                              |
| <b>Glass Cathedral</b>                                                                                                                                                                                                                                                                                                        | Your own tone sent to church: notes open a little more slowly and take their time to leave, the envelope is sent into the filter so every note dawns, and a long hall stands behind it. It is deliberately restrained — the oscillator stays in front, and only the effects that would muddy a long tail step aside. <i>Air, Halo, Shine.</i> |
| <b>Tape Séance</b>                                                                                                                                                                                                                                                                                                            | A reel that has been in the attic too long: wow and flutter bend the pitch and the echoes together, notes sag flat as they die, and there is dust on everything — hiss, crackle, a dull top end. <i>Wobble, Sag, Dust.</i>                                                                                                                    |
| <b>Insect Choir</b>                                                                                                                                                                                                                                                                                                           | The voices stop agreeing: they detune into a cluster instead of a chord, each one trembles at its own rate, and small random jumps keep the swarm moving. Arming it raises the voice count. <i>Swarm, Flutter, Skitter.</i>                                                                                                                   |
| <p><b>They are macros, not extra DSP.</b> Every character works by moving the instrument's own controls and by adding a little motion of its own, which is why arming one costs nothing you were not already paying for, why the result is visible on the panels, and why releasing it can put everything back precisely.</p> |                                                                                                                                                                                                                                                                                                                                               |

**They leave your tuning alone.** A character may spread the voices into a cluster, wow the pitch or drop a sub-octave underneath — those are the instruments. But the note you played stays the note you played: voice one sits exactly on it and the cluster grows around it, and no character rewrites *Spread*, which would re-pitch the whole stack by moving where on the photograph the extra voices read.

Seven stereo modules in a real serial chain. The order shown is the signal path: drag the :: handle, or use the arrows, and the audio graph is rewired under a short master dip so it never pops.



Every unit wears its own enclosure and its own illuminated rocker — the same switch the Spectra characters use, at pedal size. **Tube**, **Delay** and **Reverb** start switched on; the other four start off. A module that is off is removed from the audio graph entirely, so bypassing genuinely saves CPU. The rack starts in the order Tube · Stutter · Lofi · Phaser · Chorus · Delay · Reverb, and from there it is yours to arrange.

### Tube — asymmetric soft saturation

**Drive** 0–24 dB, **Tone**. A biased triode-style curve adds gentle even harmonics; four-times oversampled with DC blocking. Loudness-compensated, so driving it thickens the sound without getting louder.

### Delay — stereo echo

**Mix**, **Time** 40–900 ms, **Feedback** up to 112 %, **Offset** L↔R ±250 ms, **Character**. Past unity the loop self-oscillates into a soft limiter instead of blowing up.

### Reverb — stereo convolution

**Mix**, **Character**, **Length** 0.2–6 s. Room, Hall, Plate, **Spring** (a chain of dispersive chirps — the classic boing) and **Reverse**,

### Chorus — vintage dual-line ensemble

**Mix**, **Rate**, **Depth**. Two delay lines modulated in opposite phase, wet path rolled off dark like an old bucket-brigade unit.

### Stutter — rhythmic gate chop

**Amount**, **Rate** 1–16 Hz. A smoothed gate, so the chopping never clicks. Put it *before* delay and reverb and the tails survive the cuts.

### Lofi — crush · grit · noise · dirt

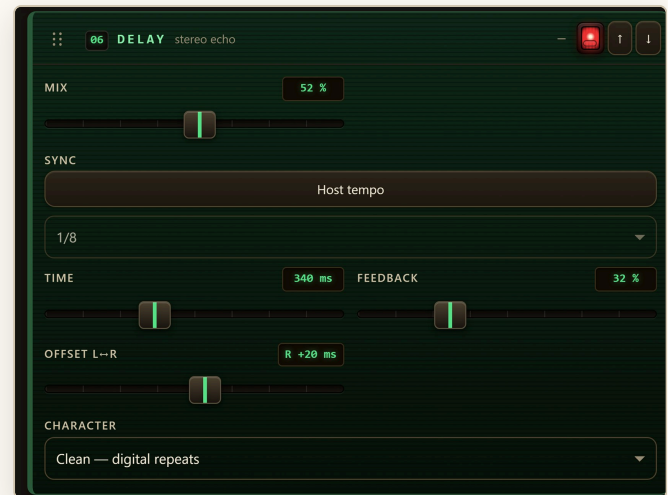
**Crush** (sample-rate reduction and bit grit), **Noise** (hiss plus sparse crackle), **Dirt** (soft normalised saturation). Everything

whose impulse swells in so notes arrive with a drawn-in breath.

## Phaser — vintage 4-stage sweep

**Mix, Rate, Depth.** Four swept all-pass stages with a little feedback: the slow hollow swoosh of a seventies stomp box.

is smoothed per sample, so the FX photo can drive it without clicks.



Tap a module's name to fold its controls open or shut.

## Delay character

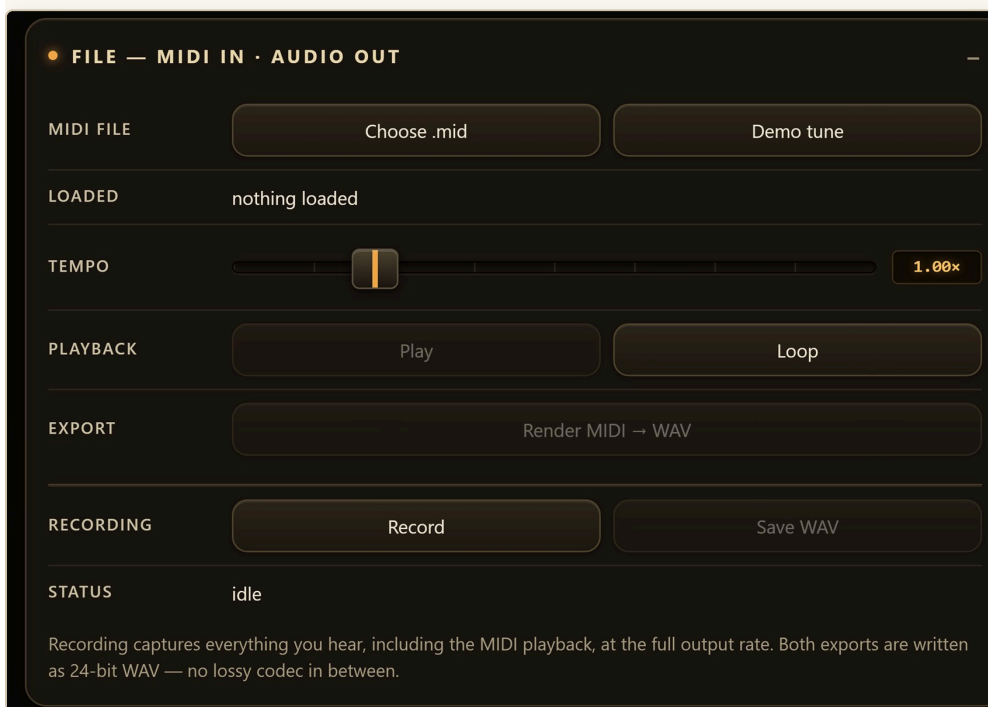
**Clean** gives digital repeats. **Tape** runs the feedback loop through gentle saturation, darker damping, a low cut and a slow wow, so every repeat gets softer, dirtier and slightly unstable — a worn analog echo. Combined with **Freeze** in the top bar, tape mode degenerates into a beautiful saturating murk you can play over.



The keyboard sets the **base note** — the note a pure red pixel plays. It adapts to the window width, and the teal marker shows the current base.

Use **- oct** and **+ oct** to move the range, and **Hold** to keep the last key down. On a computer keyboard the rows **z s x d c v g b h n j m** and **q 2 w 3 e r 5 t 6 y 7 u** play two octaves from the leftmost key. In Polyphonic voice mode, several keys — or several MIDI notes — sound together as a real chord.

**Your DAW plays it too.** MIDI from the host is handled natively and sample-accurately, with the same note allocation as the on-screen keyboard. Record a MIDI part on the track and the photos shape every note of it.



MIDI file playback, tempo scaling, offline render and the audio recorder.

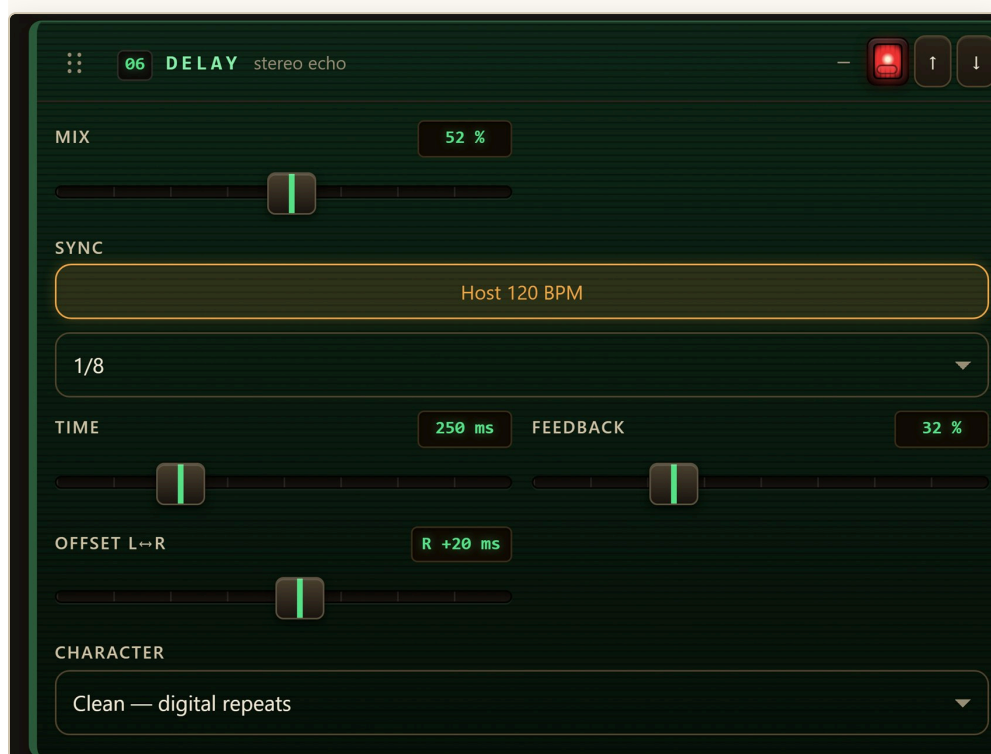
## Playing a MIDI file

**Choose .mid** imports a standard MIDI file, or **Demo tune** loads a short pentatonic phrase to try the path. Its melody drives the base note while the photos keep shaping the sound; the tempo comes from the file and **Tempo** scales it from a quarter speed to four times. In Mono/Unison mode overlapping notes are reduced to the highest one; in Polyphonic mode the file's chords play for real, up to your Notes setting.

## Recording and export

| ACTION                   | WHAT HAPPENS                                                                                                                                |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Record</b>            | Captures the plugin's output — playing, MIDI, filter sweeps and all — up to 180 seconds.                                                    |
| <b>Save WAV</b>          | Writes the capture as 24-bit stereo WAV at the session's sample rate, through a normal Windows save dialog.                                 |
| <b>Render MIDI → WAV</b> | Renders the loaded MIDI file offline, faster than real time and sample-accurate, at 48 kHz — using the colours currently under the cursors. |

## Locking the delay and the stutter to the tempo



The **Sync** switch reads the host tempo back to you; the menu beside it picks the note division.

Delay and Stutter can lock to the DAW. Press **Sync**, choose a division — 1/1 down to 1/32, dotted and triplet — and the module's own Time or Rate control is driven from the tempo the host reports, following it when you change it. The control is held while synced so a drag cannot fight it, and the switch reads back the current tempo.

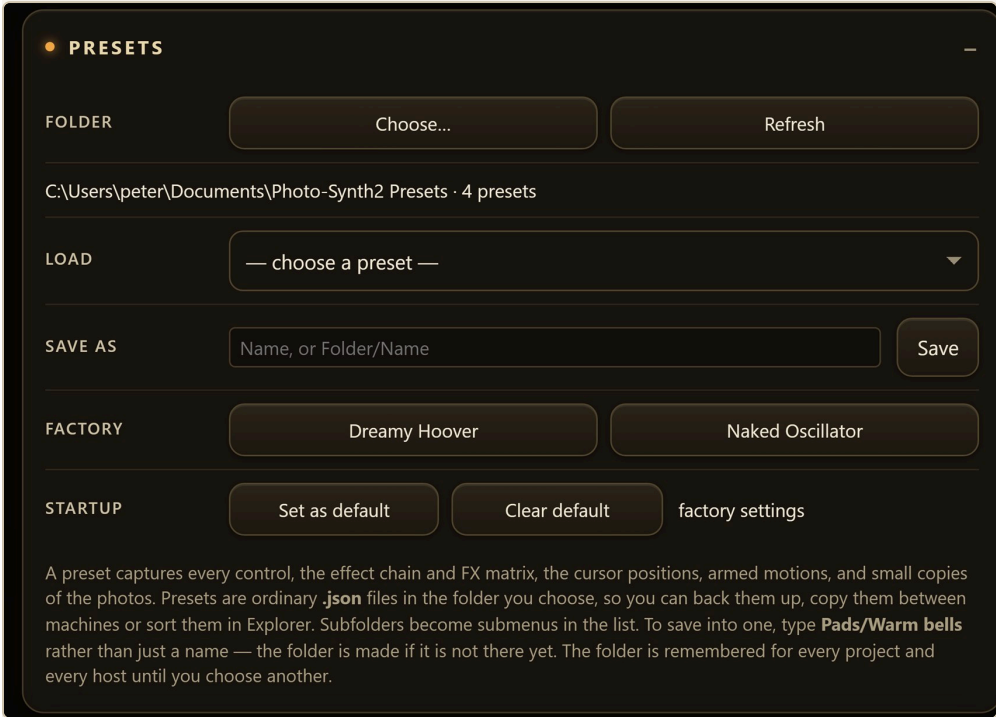
Because the page follows the tempo, a tempo change while the plugin window is closed is not tracked; the last synced value is held until you open it again.

## Factory patches and what the instrument opens on

Two patches are built in. **Dreamy Hoover** is what a new instance greets you with: seven voices stacked and gliding, pentatonic so every position on the photograph is musical, chorus and a long hall behind it, and a cursor already resting on the first photo. **Naked Oscillator** is the opposite — one voice, no effects, nothing between the picture and your ears — and it is the honest place to start building a sound of your own. Neither carries photographs, so both dress whatever pictures are loaded.

Dreamy Hoover places its cursor but does not latch it, so a new instance is silent until you touch it. Once you have a patch you would rather be greeted by, **Set as default** freezes it — controls, chain, armed characters, cursors, motions and small copies of the photographs — and new instances open on that instead. **Clear default** returns to Dreamy Hoover. A patch saved inside a DAW project always wins over both.

## Presets are files in a folder you choose



Presets keep everything, photographs included — and they are ordinary files, so they can live on a drive you back up.

A preset captures every control, the effect chain and its power switches, the FX matrix assignments, the cursor positions, the armed motions, and small copies of all four photos. It is written as an ordinary `.json` file into a folder of your choosing, so a preset library is something you can back up, copy to another machine, or sort in Explorer like any other work.

| CONTROL   | WHAT IT DOES                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Choose... | Picks the preset folder. It is remembered in the plugin's own settings — not in the session — so the same library is there in every project, every host, and after a restart. Until you pick one, the plugin uses a folder called <b>User presets</b> sitting beside the installed <code>.vst3</code> , creating it on first use; where that cannot be written to, <code>Documents\Photo-Synth2 Presets</code> is used instead. |
| Refresh   | Re-reads the folder, for when you have added or removed files behind the plugin's back.                                                                                                                                                                                                                                                                                                                                         |
| Load      | Every preset in the folder, in one menu. Subfolders become submenus, so a library can be sorted by kind — <i>Pads</i> , <i>Leads</i> , <i>Textures</i> . Choosing one loads it.                                                                                                                                                                                                                                                 |
| Save as   | Type a name and press <b>Save</b> . Type <b>Pads/Warm bells</b> instead and it is filed in that subfolder, which is created if it does not exist yet.                                                                                                                                                                                                                                                                           |

**You do not have to open this panel.** **Save** and **Load** in the title bar do the same two jobs and are reachable in Play mode. Save opens a real Save dialog on the preset folder, so you can name the file, drop it in a sub-folder, or put it somewhere else entirely — and if you do, that becomes the folder from then on. Load opens the library as a menu, sub-folders flying out sideways beside the loose presets.

Your DAW project stores the same thing independently, so reopening a session restores the sound *and* the pictures without touching the preset folder at all. The folder is for the sounds you want to reach for again; the project looks after itself.

Nine short experiments, roughly in order of adventurousness. Each one is a handful of moves — the point is to hear what the instrument does, not to build a patch.

### 1 • The first thirty seconds

Drag slowly across photo 1 in a wide arc, then leave the cursor on a bright saturated area of photo 2. Now drag on photo 1 again.

*Everything else in this manual is a variation on this.*

### 2 • Play it in tune

Sound → Pitch & scale: set **Scale** to **Minor pentatonic** and **Range** to **2 oct**. Add a little **Glide**.

*Suddenly the photo is an instrument rather than a colour meter — every colour lands on a usable note.*

### 3 • One photo, one chord

Voice mode **Mono / Unison**, **Unison voices** to 8, **Spread** to about 30 px. Put the cursor on a *busy*, detailed part of photo 1 — then on a smooth gradient.

*Detail scatters the eight voices into a chord; smooth areas collapse them into a thick unison. The texture of the picture becomes harmony.*

### 4 • Real chords from your DAW

Voice mode **Polyphonic**, **Notes** 4, **Osc / note** 2. Record a slow four-note chord progression on the track and let it play while you move the cursor on photo 1.

*The chord holds while the timbre of every note travels with your cursor.*

### 5 • A filter that plays itself

Press **Rec** on photo 2 and draw a slow circle over a colourful area, ending where you began — the white dot marks the start — then release. Set **Motion** in the top bar to about 0.5×.

*A slow endless filter sweep that follows the picture rather than an LFO — hold a note and listen to it breathe.*

### 6 • Percussive from a photograph

Latch a **deep red** spot on photo 3 (fast attack, short tail), set **Envelope** → **Filter** to about 60 %, filter type **Low**, and play staccato notes on the keyboard.

*Plucks and blips, with the filter snapping shut behind each note.*

## 7 • Metallic drone

Filter type **Comb**. Latch photo 1 on a saturated colour and photo 2 on something bright. Engage **Dark Drone**, **Sub level** up, **Cluster** around 30 ¢, **Drift** 60 %, **Drift time** long. Leave it running for a few minutes.

*A struck-string resonance that never quite repeats itself and slowly comes apart. This is what the instrument is best at.*

## 8 • Freeze and play over it

Delay **Character** to **Tape**, **Feedback** around 90 %, **Mix** up. Play a phrase, then hit **Freeze** in the top bar and keep playing.

*The phrase is held forever as a saturating tape bed while you improvise on top. Press Freeze again to let it decay.*

## 9 • Let the fourth photo run the effects

Assign the FX pad's three menus to **Lofi crush**, **Chorus mix** and **Reverb mix**, switch those three modules on, then press **Rec** on photo 4 and draw a long wandering gesture.

*The effects morph continuously along a path you drew, in combinations no three LFOs would find. Finish with **Halt**.*

## Automation

Around forty controls are exposed to the host as automatable parameters, named exactly as they are labelled in the panels — `vol`, `limiter`, `drive`, `attack`, `release`, `envAmp`, `envFlt`, `satGain`, `delayMix`, `delayTime`, `delayFeedback`, `reverbMix`, `reverbLength`, the phaser, chorus, stutter and lofi controls, and the Dark Drone set. Write automation for them as you would for any synth.

Added since version 1.0, and automatable like the rest: `tune` and `fine` (global tuning), `filterQ` (the resonance scaler, now reaching 0.05×), and `spread` now reaching 60 px. The Spectra characters, the tempo-sync switches, **Key follow** and the tuner's Match button are panel controls rather than parameters — they move the automatable sliders on your behalf, or are stored with the patch, so what they do is recorded in the project and visible on the panels.

## What the project remembers

Saving your session stores every control, the effect order and power switches, the FX matrix, the cursor positions, the armed motions and compressed copies of all four photographs. Reopening the session restores the instrument exactly, pictures included, with no external files to keep track of.

## Two things worth knowing

- **The window may be left open for gestures.** The engine keeps sounding with the editor closed and host MIDI plays normally, but the parts that are genuinely driven by the interface — looping motions, the Dark Drone drift walk, the FX photo, and re-sampling of the photos — need the plugin window open to run. For a fixed sound, close it freely.
- **Resonance plus drive can peak.** That is the nature of a ladder filter with the envelope sweeping it, and it is why **Limit** exists. It is on at 40 % by default and transparent at that setting.
- **The offline MIDI render takes the filter as a snapshot.** Key follow, the photo drift and the Spectra modulations do not track note by note there, exactly as they did not before. Live playing and the audio recorder follow normally.

## Photo-Synth 2

This VST comes from the curved mind of **Professor Brokild**, and was programmed with VScode, ChatGPT and Claude. Use of this is on your own peril. It may generate unique, beautiful, scary or useless sounds for you, and it may crash your DAW project. Have fun, play unsafely, unwisely, and unboringly.

Cheers, Peter Bøggild, August 2026.

BUILD 260821.14

Close

The **About** button, in the top bar.

**Photo-Synth 2** · Brokild Apps · August 2026. Native VST3 and standalone port of the browser instrument Photo Synth. The interface is the original page; the audio engine — voice pool, ladder filter, effects rack, recorder and offline render — is native C++.

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